WORKED SOLUTION

STEP I 2013, Question 2

Pure • Floor Function Equations

STEP I 2013, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q2

The key idea

The function $\lfloor x \rfloor$ is constant on each interval $[k, k+1)$, so on that interval $f(x) = k/x$ is just a piece of a rectangular hyperbola. The whole graph is a string of these pieces, one for each integer k. Solving $f(x) = c$ then comes down to asking which pieces reach height c. On the piece for k the only candidate is $x = k/c$, and it counts precisely when it lands in $[k, k+1)$.

Solution

Part (i)

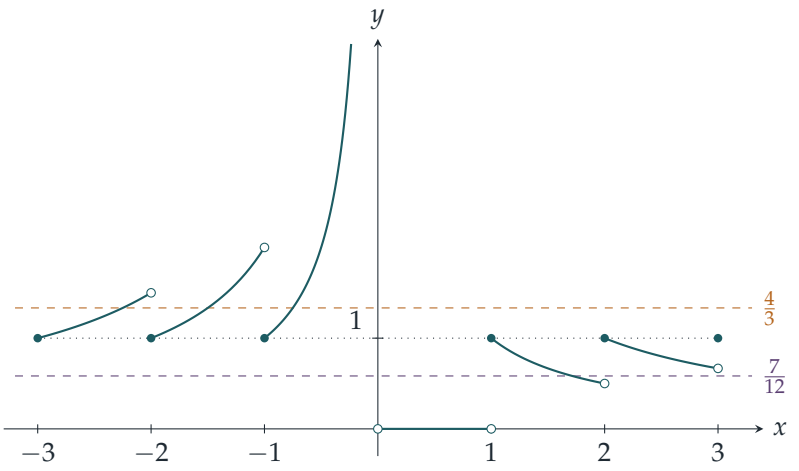
Fix an integer k and look at $k \le x < k+1$ (with $x \neq 0$), where $f(x) = k/x$.

- $k = 0$, i.e. $0 < x < 1$: here $f(x) = 0$.
- $k \ge 1$: $f(x) = k/x$ decreases from $f(k) = 1$ towards $k/(k+1)$, which it never reaches. So this piece takes exactly the values in $\left(\frac{k}{k+1}, 1\right]$.
- $k \le -1$: write $k = -m$ with $m \ge 1$. Then x and k are both negative, so $f(x) = m/|x| > 0$. As x increases from $-m$ towards $-m+1$, $|x|$ decreases, so f increases from $f(-m) = 1$. For $m \ge 2$ it tends to $\frac{m}{m-1}$, which is never reached, so this piece takes exactly the values in $\left[1, \frac{m}{m-1}\right)$. For $m = 1$ we have $f(x) = -1/x \to +\infty$ as $x \to 0^-$, so this piece takes the values in $[1, \infty)$.

For $-3 \le x \le 3$ the pieces are

x	$[-3,-2)$	$[-2,-1)$	$[-1,0)$	$(0,1)$	$[1,2)$	$[2,3)$
$f(x)$	$-3/x$	$-2/x$	$-1/x$	0	$1/x$	$2/x$
values	$[1, \frac{3}{2})$	$[1,2)$	$[1,\infty)$	0	$(\frac{1}{2}, 1]$	$(\frac{2}{3}, 1]$

and finally $f(3) = 1$. Every piece starts at height 1 at its left-hand (integer) end, where the point is included, and has an open end on the right.



Part (ii)

On the piece where $\lfloor x \rfloor = k$, the equation $f(x) = c$ (with $c \neq 0$) says $k/x = c$, so $x = k/c$. This is a genuine root exactly when $k \leq k/c < k+1$. So for each c we only have to decide which pieces reach height c, using the ranges from part (i).

$f(x) = \frac{7}{12}$. Since $0 < \frac{7}{12} < 1$, only pieces with $k \geq 1$ can help, and we need $\frac{k}{k+1} < \frac{7}{12}$, i.e. $12k < 7k + 7$, i.e. $k < \frac{7}{5}$. So only $k = 1$ works, giving $x = 1/\frac{7}{12}$:

$$x = \tfrac{12}{7}.$$

On the graph, the line $y = \frac{7}{12}$ meets only the piece over $[1, 2)$, since the piece over $[2, 3)$ stays above $\frac{2}{3}$.

$f(x) = \frac{17}{24}$. Now we need $\frac{k}{k+1} < \frac{17}{24}$, i.e. $24k < 17k + 17$, i.e. $k < \frac{17}{7}$, so $k = 1$ or $k = 2$. The roots are $x = k/\frac{17}{24}$:

$$x = \tfrac{24}{17} \quad \text{and} \quad x = \tfrac{48}{17}.$$

(Indeed $2 \leq \frac{48}{17} \approx 2.82 < 3$.)

$f(x) = \frac{4}{3}$. Since $\frac{4}{3} > 1$, only the negative pieces can help. The piece with $k = -m$ reaches $\frac{4}{3}$ when $\frac{4}{3} < \frac{m}{m-1}$ (or $m = 1$), i.e. $4m - 4 < 3m$, i.e. $m < 4$. So $m = 1, 2, 3$, and the roots are $x = -m/\frac{4}{3} = -\frac{3m}{4}$:

$$x = -\tfrac{3}{4}, \quad x = -\tfrac{3}{2}, \quad x = -\tfrac{9}{4}.$$

The case $m = 4$ just fails: the piece over $[-4, -3)$ only gets up to values below $\frac{4}{3}$.

Part (iii)

Here $0 < \frac{9}{10} < 1$, so the roots come from the pieces with $k \geq 1$ and $\frac{k}{k+1} < \frac{9}{10}$, i.e. $10k < 9k + 9$, i.e. $k < 9$. So $k = 1, 2, \ldots, 8$ each give a root $x = \frac{10k}{9}$. The largest is the one with $k = 8$:

$$x = \tfrac{80}{9} \approx 8.89,$$

which does lie in $[8, 9)$.

The final part

First, the values of c that give no roots or infinitely many.

- $c < 0$: no roots, because $f(x) \geq 0$ for $x > 0$ and $f(x) > 0$ for $x < 0$.
- $c = 0$: every $x \in (0, 1)$ is a root, so there are infinitely many.
- $c = 1$: every nonzero integer is a root, so again infinitely many.

Ask yourself How do we count roots for a general c, without drawing the graph?

Use the ranges from part (i): each piece contributes at most one root, so we just count the pieces whose range contains c.

Case $0 < c < 1$**.** The pieces that reach c are those with $k \geq 1$ and $\frac{k}{k+1} < c$. Since $\frac{k}{k+1}$ increases with k, these are $k = 1, 2, \ldots$ up to the last k with $\frac{k}{k+1} < c$. So there are exactly n roots precisely when the nth piece reaches c but the $(n+1)$th does not:

$$\frac{n}{n+1} < c \leq \frac{n+1}{n+2}.$$

(For $0 < c \leq \frac{1}{2}$ no piece reaches c, and there are no roots.)

Case $c > 1$**.** The pieces that reach c are those with $k = -m$, where either $m = 1$ or $\frac{m}{m-1} > c$. Since $\frac{m}{m-1} = 1 + \frac{1}{m-1}$ decreases as m increases, these are $m = 1, 2, \ldots$ up to the last m with $\frac{m}{m-1} > c$. So there are exactly n roots precisely when the piece for $m = n$ reaches c but the piece for $m = n + 1$ does not:

$$\frac{n+1}{n} \leq c < \frac{n}{n-1} \qquad (n \geq 2),$$

and, for $n = 1$, simply $c \geq 2$ (the piece over $[-1, 0)$ always reaches c, and the piece over $[-2, -1)$ doesn't when $c \geq 2$).

Answer. For $n \geq 1$, the equation $f(x) = c$ has exactly n roots if and only if

$$\frac{n}{n+1} < c \leq \frac{n+1}{n+2} \qquad \text{or} \qquad \frac{n+1}{n} \leq c < \frac{n}{n-1},$$

where for $n = 1$ the second condition is read as $c \geq 2$. Notice where the equality signs go. Both come from open ends of pieces: $c = \frac{n+1}{n+2}$ is not reached by the piece over $[n+1, n+2)$, and $c = \frac{n+1}{n}$ is not reached by the piece over $[-(n+1), -n)$. So both of these boundary values still give exactly n roots.

Checks

- Substitute back: $\lfloor \frac{12}{7} \rfloor = 1$ and $1/\frac{12}{7} = \frac{7}{12}$; $\lfloor \frac{48}{17} \rfloor = 2$ and $2/\frac{48}{17} = \frac{17}{24}$; $\lfloor -\frac{9}{4} \rfloor = -3$ and $-3/(-\frac{9}{4}) = \frac{4}{3}$; $\lfloor \frac{80}{9} \rfloor = 8$ and $8/\frac{80}{9} = \frac{9}{10}$.
- The general answer agrees with the specific cases: $\frac{2}{3} < \frac{17}{24} \leq \frac{3}{4}$ gives $n = 2$; $\frac{4}{3} \leq \frac{4}{3} < \frac{3}{2}$ gives $n = 3$; $\frac{8}{9} < \frac{9}{10} \leq \frac{9}{10}$ gives $n = 8$. The last two sit exactly on an endpoint of the range, which tests the placing of $\leq$ and $<$.
- A boundary value: $c = 2$ should give one root. Indeed $x = -\frac{1}{2}$ works, while $x = -1$ (from $k = -2$) has $\lfloor -1 \rfloor = -1 \neq -2$.
- Try $c = \frac{1}{2}$: the conditions give no n, and indeed $x = k/c = 2k$ never satisfies $\lfloor 2k \rfloor = k$ for $k \neq 0$.

Common slips

- Forgetting that $\lfloor x \rfloor$ rounds *down* for negative x, so that $\lfloor -2.25 \rfloor = -3$, not -2. This loses the negative branch for $c > 1$ entirely.
- Getting the endpoints wrong: each piece includes its left-hand (integer) end, at height 1, but not its right-hand end. This decides whether $\leq$ or $<$ appears in the final conditions.
- Leaving out $c = 0$ and $c = 1$, where there are infinitely many roots, or $c < 0$ and $0 < c \leq \frac{1}{2}$, where there are none.

Beyond STEP

Functions such as $\lfloor x \rfloor / x$ are the simplest examples of functions that are continuous on pieces but jump at isolated points. Being careful about which endpoints are attained is exactly the habit you need for suprema and infima in a first analysis course. Here $\sup f$ over $x > 0$ is 1 and is attained, while the infimum over $x \geq 1$ is $\frac{1}{2}$ and is not. The same function also shows that $\lfloor x \rfloor / x \to 1$ as $x \to \infty$, a first taste of the estimates used in analytic number theory. Abbott's *Understanding Analysis* is a good place to read more about suprema, limits and continuity.

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